Environment Variable and Set-UID Program Lab

This lab looks at how environment variables affect the way programs run. Environment variables are a set of dynamic named values that can affect the way running processes will behave on a computer, but if a programmer doesn't realize they're being used, it can cause security issues. In this lab, we explore how these variables get passed between processes and how they can change program behavior, especially in programs with higher privileges like Set-UID programs.

Task 1: Manipulating Environment Variables

You can use the **printenv** or **env** command to see the full list of your environment variables

```
deven@ubuntu:~/lab1$ printenv PWD
/home/deven/lab1
deven@ubuntu:~/lab1$ env | grep PWD
PWD=/home/deven/lab1
OLDPWD=/home/deven
```

The commands to see environment specific variables are as shown. As you can see my variable titled PWD is in the lab1 directory. To make a new environment variable you can use **export** and to remove it use **unset**:

```
deven@ubuntu:~/lab1$ export TEST=ThisisanEnvVar
deven@ubuntu:~/lab1$ printenv TEST
ThisisanEnvVar
deven@ubuntu:~/lab1$ unset TEST
deven@ubuntu:~/lab1$ printenv TEST
```

Task 2: Passing Environment Variables from Parent Process to Child Process

For this task we will compare how a child process inherits environment variables from a parent process. For this we compile a program with a parent and child process, getting the outputs of environment variables from each to see the difference. When comparing using **diff** my terminal picked up on no differences between the parent and child case as shown. After changing the code of the program and recompiling no significant change was found.

```
deven@ubuntu:~/lab1/Labsetup$ nano myprintenv.c
deven@ubuntu:~/lab1/Labsetup$ gcc myprintenv.c
deven@ubuntu:~/lab1/Labsetup$ ./a.out > output1.txt
deven@ubuntu:~/lab1/Labsetup$ diff output1.txt output2.txt
deven@ubuntu:~/lab1/Labsetup$
```

Here’s the parent output

```
13 DESKTOP_SESSION=ubuntu
14 GTK_MODULES=gail:atk-bridge
15 PWD=/home/deven/lab1/Labsetup
16 LOGNAME=deven
17 XDG_SESSION_DESKTOP=ubuntu
18 XDG_SESSION_TYPE=x11
19 GPG_AGENT_INFO=/run/user/1003/gnupg/S.gpg-agent:0:1
20 SYSTEMD_EXEC_PID=13970
21 XAUTHORITY=/run/user/1003/gdm/Xauthority
22 WINDOWPATH=3
23 HOME=/home/deven
24 USERNAME=deven
25 IM_CONFIG_PHASE=1
26 LANG=en_US.UTF-8
27 LS_COLORS=rs=0:di=01;34:ln=01;36:mh=00:pi=40;33:so=01;35:do=01;35:bd=40;33;01:cd=40;33;01:or=40;31;01:mi=00:su=37;
   7z=01;31:*.rz=01;31:*.cab=01;31:*.wim=01;31:*.swm=01;31:*.dwm=01;31:*.esd=01;31:*.jpg=01;35:*.jpeg=01;35:*.mjpg=01
28 XDG_CURRENT_DESKTOP=ubuntu:GNOME
29 VTE_VERSION=6800
30 GNOME_TERMINAL_SCREEN=/org/gnome/Terminal/screen/1d8a01b1_d45a_46d5_9a59_1b748507085a
31 LESSCLOSE=/usr/bin/lesspipe %s %s
32 XDG_SESSION_CLASS=user
33 TERM=xterm-256color
34 LESSOPEN=| /usr/bin/lesspipe %s
35 USER=deven
36 GNOME_TERMINAL_SERVICE=:1.95
37 DISPLAY=:2
38 SHLVL=1
39 QT_IM_MODULE=ibus
40 XDG_RUNTIME_DIR=/run/user/1003
41 XDG_DATA_DIRS=/usr/share/ubuntu:/usr/share/gnome:/usr/local/share/:/usr/share/:/var/lib/snapd/desktop
42 PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin:/snap/bin
:43 GDMSESSION=ubuntu
```

And the child output:

```
13 DESKTOP_SESSION=ubuntu
14 GTK_MODULES=gail:atk-bridge
15 PWD=/home/deven/lab1/Labsetup
16 LOGNAME=deven
17 XDG_SESSION_DESKTOP=ubuntu
18 XDG_SESSION_TYPE=x11
19 GPG_AGENT_INFO=/run/user/1003/gnupg/S.gpg-agent:0:1
20 SYSTEMD_EXEC_PID=13970
21 XAUTHORITY=/run/user/1003/gdm/Xauthority
22 WINDOWPATH=3
23 HOME=/home/deven
24 USERNAME=deven
25 IM_CONFIG_PHASE=1
26 LANG=en_US.UTF-8
27 LS_COLORS=rs=0:di=01;34:ln=01;36:mh=00:pi=40;33:so=01;35:do=01;35:bd=40;33;01:cd=40;33;01:or=40;31;01:mi=00:su=37;
   7z=01;31:*.rz=01;31:*.cab=01;31:*.wim=01;31:*.swm=01;31:*.dwm=01;31:*.esd=01;31:*.jpg=01;35:*.jpeg=01;35:*.mjpg=01
28 XDG_CURRENT_DESKTOP=ubuntu:GNOME
29 VTE_VERSION=6800
30 GNOME_TERMINAL_SCREEN=/org/gnome/Terminal/screen/1d8a01b1_d45a_46d5_9a59_1b748507085a
31 LESSCLOSE=/usr/bin/lesspipe %s %s
32 XDG_SESSION_CLASS=user
33 TERM=xterm-256color
34 LESSOPEN=| /usr/bin/lesspipe %s
35 USER=deven
36 GNOME_TERMINAL_SERVICE=:1.95
37 DISPLAY=:2
38 SHLVL=1
39 QT_IM_MODULE=ibus
40 XDG_RUNTIME_DIR=/run/user/1003
41 XDG_DATA_DIRS=/usr/share/ubuntu:/usr/share/gnome:/usr/local/share/:/usr/share/:/var/lib/snapd/desktop
42 PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin:/snap/bin
43 GDMSESSION=ubuntu
```

Task 3: Environment Variable and execve()

In this section we will be testing how environment variables are affected by when a new program is executed. We will compile and run a function **execve()** which will introduce a new program to the environment and we can observe how the variables are affected. When running an execve() program with NULL as the environment parameter we get no output:

```
deven@ubuntu:~/lab1/Labsetup$ gcc myenv.c -o myenv
deven@ubuntu:~/lab1/Labsetup$ ./myenv
deven@ubuntu:~/lab1/Labsetup$
```

However, if we change NULL to environ the function then has an environment to reference and prints out the variables.:

```
deven@ubuntu:~/lab1/Labsetup$ nano myenv.c
deven@ubuntu:~/lab1/Labsetup$ gcc myenv.c -o myenv
deven@ubuntu:~/lab1/Labsetup$ ./myenv
SHELL=/bin/bash
COLORTERM=truecolor
SUDO_GID=1004
SUDO_COMMAND=/usr/bin/su root
SUDO_USER=deven
PWD=/home/deven/lab1/Labsetup
LOGNAME=deven
XAUTHORITY=/run/user/1003/gdm/Xauthority
HOME=/home/deven
LANG=en_US.UTF-8
LS_COLORS=rs=0:di=01;34:ln=01;36:mh=00:pi=40;33:so=01;35:do=01;35:bd=40;33;01:cd=40;33;01:or=40;31;01:mi
41:tw=30;42:ow=34;42:st=37;44:ex=01;32:*.tar=01;31:*.tgz=01;31:*.arc=01;31:*.arj=01;31:*.taz=01;31:*.lha
31:*.lzma=01;31:*.tlz=01;31:*.txz=01;31:*.tzo=01;31:*.t7z=01;31:*.zip=01;31:*.z=01;31:*.dz=01;31:*.gz=01
.lzo=01;31:*.xz=01;31:*.zst=01;31:*.tzst=01;31:*.bz2=01;31:*.bz=01;31:*.tbz=01;31:*.tbz2=01;31:*.tz=01;3
jar=01;31:*.war=01;31:*.ear=01;31:*.sar=01;31:*.rar=01;31:*.alz=01;31:*.ace=01;31:*.zoo=01;31:*.cpio=01;
ab=01;31:*.wim=01;31:*.swm=01;31:*.dwm=01;31:*.esd=01;31:*.jpg=01;35:*.jpeg=01;35:*.mjpg=01;35:*.mjpeg=0
:*.pbm=01;35:*.pgm=01;35:*.ppm=01;35:*.tga=01;35:*.xbm=01;35:*.xpm=01;35:*.tif=01;35:*.tiff=01;35:*.png=
35:*.mng=01;35:*.pcx=01;35:*.mov=01;35:*.mpg=01;35:*.mpeg=01;35:*.m2v=01;35:*.mkv=01;35:*.webm=01;35:*.w
01;35:*.m4v=01;35:*.mp4v=01;35:*.vob=01;35:*.qt=01;35:*.nuv=01;35:*.wmv=01;35:*.asf=01;35:*.rm=01;35:*.r
01;35:*.fli=01;35:*.flv=01;35:*.gl=01;35:*.dl=01;35:*.xcf=01;35:*.xwd=01;35:*.yuv=01;35:*.cgm=01;35:*.em
;35:*.aac=00;36:*.au=00;36:*.flac=00;36:*.m4a=00;36:*.mid=00;36:*.midi=00;36:*.mka=00;36:*.mp3=00;36:*.m
;36:*.wav=00;36:*.oga=00;36:*.opus=00;36:*.spx=00;36:*.xspf=00;36:
XDG_CURRENT_DESKTOP=ubuntu:GNOME
LESSCLOSE=/usr/bin/lesspipe %s %s
TERM=xterm-256color
LESSOPEN=| /usr/bin/lesspipe %s
USER=deven
DISPLAY=:2
SHLVL=2
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin
SUDO_UID=1003
MAIL=/var/mail/deven
_=./myenv
deven@ubuntu:~/lab1/Labsetup$
```

Task 4: Environment Variables and system()

The system() function works similarly to execve() however it doesn't directly execute a command. We see that ultimately we end up with the same result when using **system("/usr/bin/env");**

```
deven@ubuntu:~/lab1/Labsetup$ nano mysystem.c
deven@ubuntu:~/lab1/Labsetup$ gcc mysystem.c -o mysystem
deven@ubuntu:~/lab1/Labsetup$ ./mysystem
SUDO_GID=1004
LESSOPEN=| /usr/bin/lesspipe %s
MAIL=/var/mail/deven
USER=deven
SHLVL=2
HOME=/home/deven
COLORTERM=truecolor
SUDO_UID=1003
LOGNAME=deven
_=./mysystem
TERM=xterm-256color
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin
DISPLAY=:2
LANG=en_US.UTF-8
XDG_CURRENT_DESKTOP=ubuntu:GNOME
XAUTHORITY=/run/user/1003/gdm/Xauthority
LS_COLORS=rs=0:di=01;34:ln=01;36:mh=00:pi=40;33:so=01;35:do=01;35:bd=40;33;01:cd=40;33;01:or=40;31;01:mi=00:su=37;41:sg=30;43:ca=30;
41:tw=30;42:ow=34;42:st=37;44:ex=01;32:*.tar=01;31:*.tgz=01;31:*.arc=01;31:*.arj=01;31:*.taz=01;31:*.lha=01;31:*.lz4=01;31:*.lzh=01;
31:*.lzma=01;31:*.tlz=01;31:*.txz=01;31:*.tzo=01;31:*.t7z=01;31:*.zip=01;31:*.z=01;31:*.dz=01;31:*.gz=01;31:*.lrz=01;31:*.lz=01;31:*
.lzo=01;31:*.xz=01;31:*.zst=01;31:*.tzst=01;31:*.bz2=01;31:*.bz=01;31:*.tbz=01;31:*.tbz2=01;31:*.tz=01;31:*.deb=01;31:*.rpm=01;31:*.
jar=01;31:*.war=01;31:*.ear=01;31:*.sar=01;31:*.rar=01;31:*.alz=01;31:*.ace=01;31:*.zoo=01;31:*.cpio=01;31:*.7z=01;31:*.rz=01;31:*.c
ab=01;31:*.wim=01;31:*.swm=01;31:*.dwm=01;31:*.esd=01;31:*.jpg=01;35:*.jpeg=01;35:*.mjpg=01;35:*.mjpeg=01;35:*.gif=01;35:*.bmp=01;35
:*.pbm=01;35:*.pgm=01;35:*.ppm=01;35:*.tga=01;35:*.xbm=01;35:*.xpm=01;35:*.tif=01;35:*.tiff=01;35:*.png=01;35:*.svg=01;35:*.svgz=01;
35:*.mng=01;35:*.pcx=01;35:*.mov=01;35:*.mpg=01;35:*.mpeg=01;35:*.m2v=01;35:*.mkv=01;35:*.webm=01;35:*.webp=01;35:*.ogm=01;35:*.mp4=
01;35:*.m4v=01;35:*.mp4v=01;35:*.vob=01;35:*.qt=01;35:*.nuv=01;35:*.wmv=01;35:*.asf=01;35:*.rm=01;35:*.rmvb=01;35:*.flc=01;35:*.avi=
01;35:*.fli=01;35:*.flv=01;35:*.gl=01;35:*.dl=01;35:*.xcf=01;35:*.xwd=01;35:*.yuv=01;35:*.cgm=01;35:*.emf=01;35:*.ogv=01;35:*.ogx=01
;35:*.aac=00;36:*.au=00;36:*.flac=00;36:*.m4a=00;36:*.mid=00;36:*.midi=00;36:*.mka=00;36:*.mp3=00;36:*.mpc=00;36:*.ogg=00;36:*.ra=00
;36:*.wav=00;36:*.oga=00;36:*.opus=00;36:*.spx=00;36:*.xspf=00;36:
SUDO_COMMAND=/usr/bin/su root
SHELL=/bin/bash
SUDO_USER=deven
LESSCLOSE=/usr/bin/lesspipe %s %s
PWD=/home/deven/lab1/Labsetup
```

Task 5: Environment Variable and Set-UID Programs

Set-UID programs allow programs to take on the same privileges the owner of the program has. It can be risky due to this fact. We will create a program that prints environment variables and change the ownership to root. This way a user who has access to the program can use it with root permissions. As a normal user we can set environment variables. I set PATH LD_LIBRARY_PATH and SUS. The only environment variable that did not print is LD_LIBRARY_PATH this is for security to keep the libraries from being tampered with.

```
deven@ubuntu:~/lab1/Labsetup$ nano foo.c
deven@ubuntu:~/lab1/Labsetup$ gcc foo.c -o foo
deven@ubuntu:~/lab1/Labsetup$ sudo chown root foo
[sudo] password for deven:
deven@ubuntu:~/lab1/Labsetup$ sudo chmod 4755 foo
deven@ubuntu:~/lab1/Labsetup$ export PATH
deven@ubuntu:~/lab1/Labsetup$ export LD_LIBRARY_PATH
deven@ubuntu:~/lab1/Labsetup$ export SUS=ThisIsASusVariable
deven@ubuntu:~/lab1/Labsetup$ ./foo
SHELL=/bin/bash
COLORTERM=truecolor
SUDO_GID=1004
SUDO_COMMAND=/usr/bin/su root
SUDO_USER=deven
PWD=/home/deven/lab1/Labsetup
LOGNAME=deven
SUS=ThisIsASusVariable
XAUTHORITY=/run/user/1003/gdm/Xauthority
HOME=/home/deven
LANG=en_US.UTF-8
LS_COLORS=rs=0:di=01;34:ln=01;36:mh=00:pi=40;33:so=01;35:do=
41:tw=30;42:ow=34;42:st=37;44:ex=01;32:*.tar=01;31:*.tgz=01;
31:*.lzma=01;31:*.tlz=01;31:*.txz=01;31:*.tzo=01;31:*.t7z=01
.lzo=01;31:*.xz=01;31:*.zst=01;31:*.tzst=01;31:*.bz2=01;31:*
jar=01;31:*.war=01;31:*.ear=01;31:*.sar=01;31:*.rar=01;31:*.
ab=01;31:*.wim=01;31:*.swm=01;31:*.dwm=01;31:*.esd=01;31:*.j
:*.pbm=01;35:*.pgm=01;35:*.ppm=01;35:*.tga=01;35:*.xbm=01;35
35:*.mng=01;35:*.pcx=01;35:*.mov=01;35:*.mpg=01;35:*.mpeg=01
01;35:*.m4v=01;35:*.mp4v=01;35:*.vob=01;35:*.qt=01;35:*.nuv=
01;35:*.fli=01;35:*.flv=01;35:*.gl=01;35:*.dl=01;35:*.xcf=01
;35:*.aac=00;36:*.au=00;36:*.flac=00;36:*.m4a=00;36:*.mid=00
;36:*.wav=00;36:*.oga=00;36:*.opus=00;36:*.spx=00;36:*.xspf=
XDG_CURRENT_DESKTOP=ubuntu:GNOME
LESSCLOSE=/usr/bin/lesspipe %s %s
TERM=xterm-256color
LESSOPEN=| /usr/bin/lesspipe %s
USER=deven
DISPLAY=:2
```

Task 6: The PATH Environment Variable and Set-UID Programs.

A malicioous user who has access to a Set-UID program can change the PATH variable. This is a security risk. We will demonstrate how this can be taken advantage of. We created a program with the vulnerability of having **system("ls")** which works off of relative path. This way a bad actor could put a malicious program in any directory and change the PATH enviornment variable to that directory to execute their program. Here I created the program and the shell warned me of the risk:

```
deven@ubuntu:~/lab1/Labsetup$ nano victim.c
deven@ubuntu:~/lab1/Labsetup$ export PATH=/home/seed:$PATH
deven@ubuntu:~/lab1/Labsetup$ gcc victim.c -o victim
victim.c: In function 'main':
victim.c:3:1: warning: implicit declaration of function 'system' [-Wimplicit-function-declaration]
    3 | system("ls");
      | ^~~~~~
deven@ubuntu:~/lab1/Labsetup$ sudo chown root victim
[sudo] password for deven:
```

I then made a malicious program in a fake directory

```
deven@ubuntu:~$ cd Desktop
deven@ubuntu:~/Desktop$ mkdir seed
deven@ubuntu:~/Desktop$ chmod u+w+x seed
deven@ubuntu:~/Desktop$ cd /seed
bash: cd: /seed: No such file or directory
deven@ubuntu:~/Desktop$ cd seed
deven@ubuntu:~/Desktop/seed$ nano ls
deven@ubuntu:~/Desktop/seed$ cd ~
```

Now I'll change the PATH variable and run the program.

```
deven@ubuntu:~/lab1/Labsetup$ export PATH=~/Desktop/seed:$PATH
deven@ubuntu:~/lab1/Labsetup$ printenv PATH
/home/deven/Desktop/seed:/Desktop/seed:/usr/local/sbin:/usr/local/bin:/usr/sbin:/
p/bin:/snap/bin
deven@ubuntu:~/lab1/Labsetup$ ./victim
You've been hacked! XD
deven@ubuntu:~/lab1/Labsetup$
```

The malicious program was ran and printed off You've been hacked.

Task 7: The LD_PRELOAD Environment Variable and Set-UID Programs

In this task, we will explore how the LD_LIBRARY_PATH affects the dynamic linker/loader. First we'll make a program that overrides the sleep() function in lib and compile it to a dynamic linker

```
deven@ubuntu:~/lab1/Labsetup$ nano mylib.c
deven@ubuntu:~/lab1/Labsetup$ gcc -fPIC -g -c mylib.c
deven@ubuntu:~/lab1/Labsetup$ gcc -shared -o libmylib.so.1.0.1 mylib.o -lc
deven@ubuntu:~/lab1/Labsetup$ export LD_PRELOAD=./libmylib.so.1.0.1
```

Then we will do the same for another program called myprog.c when running this program we expect the sleep function to be disrupted and have a statement printed saying so.

When running the program normally and as set_UID as a normal user nothing happened the sleep functioin was not disrupted W

```
deven@ubuntu:~/lab1/Labsetup$ nano myprog.c
ERROR: ld.so: object './libmylib.so.1.0.1' from LD_PRELOAD cannot be preloaded (cannot open shared object file):
deven@ubuntu:~/lab1/Labsetup$ gcc -fPIC -g -c myprog.c
ERROR: ld.so: object './libmylib.so.1.0.1' from LD_PRELOAD cannot be preloaded (cannot open shared object file):
ERROR: ld.so: object './libmylib.so.1.0.1' from LD_PRELOAD cannot be preloaded (cannot open shared object file):
ERROR: ld.so: object './libmylib.so.1.0.1' from LD_PRELOAD cannot be preloaded (cannot open shared object file):
deven@ubuntu:~/lab1/Labsetup$ gcc -shared -o libmylib.so.1.0.1 myprog.o -lc
ERROR: ld.so: object './libmylib.so.1.0.1' from LD_PRELOAD cannot be preloaded (cannot open shared object file):
ERROR: ld.so: object './libmylib.so.1.0.1' from LD_PRELOAD cannot be preloaded (cannot open shared object file):
ERROR: ld.so: object './libmylib.so.1.0.1' from LD_PRELOAD cannot be preloaded (cannot open shared object file):
deven@ubuntu:~/lab1/Labsetup$ export LD_PRELOAD=$PWD/libmylib.so.1.0.1
deven@ubuntu:~/lab1/Labsetup$ nano myprog.c
deven@ubuntu:~/lab1/Labsetup$ gcc -fPIC -g -c myprog.c
deven@ubuntu:~/lab1/Labsetup$ gcc -shared -o libmylib.so.1.0.1 myprog.o -lc
deven@ubuntu:~/lab1/Labsetup$ ./myprog
bash: ./myprog: No such file or directory
deven@ubuntu:~/lab1/Labsetup$ gcc myprog.c -o myprog
deven@ubuntu:~/lab1/Labsetup$ ./myprog
deven@ubuntu:~/lab1/Labsetup$ sudo chown root myprog
[sudo] password for deven:
deven@ubuntu:~/lab1/Labsetup$ chmod 4755 myprog
chmod: changing permissions of 'myprog': Operation not permitted
deven@ubuntu:~/lab1/Labsetup$ sudo su root
root@ubuntu:/home/deven/lab1/Labsetup# chmod 4755 myprog
root@ubuntu:/home/deven/lab1/Labsetup# sudo su deven
deven@ubuntu:~/lab1/Labsetup$ myprog
myprog: command not found
deven@ubuntu:~/lab1/Labsetup$ ./myprog
```

While trying to run the program as root and as a different user, I still got no results. Perhaps there is an error in my method.

```
deven@ubuntu:~/lab1/Labsetup$ sudo su root
[sudo] password for deven:
Sorry, try again.
[sudo] password for deven:
root@ubuntu:/home/deven/lab1/Labsetup# export LD_PRELOAD=./libmylib.so.1.0.1
root@ubuntu:/home/deven/lab1/Labsetup# ./myprog
root@ubuntu:/home/deven/lab1/Labsetup# sudo su sally
sally@ubuntu:/home/deven/lab1/Labsetup$ export LD_PRELOAD=./libmylib.so.1.0.1
sally@ubuntu:/home/deven/lab1/Labsetup$ ./myprog
sally@ubuntu:/home/deven/lab1/Labsetup$
```

Task 8: Invoking External Programs Using system() versus execve()

We create a program that uses system(command) to run a command. The user can enter in a command to the terminal with this method because system is vulnerable and the program has root privliges.

```
even@ubuntu:~/lab1/Labsetup$ gcc catall.c -o catall
even@ubuntu:~/lab1/Labsetup$ sudo chown root catall
[sudo] password for deven:
orry, try again.
[sudo] password for deven:
even@ubuntu:~/lab1/Labsetup$ chmod 4755 catall
hmod: changing permissions of 'catall': Operation not permitted
even@ubuntu:~/lab1/Labsetup$ sudo su root
oot@ubuntu:/home/deven/lab1/Labsetup# chmod 4755 catall
oot@ubuntu:/home/deven/lab1/Labsetup# sudo su deven
even@ubuntu:~/lab1/Labsetup$ sudo su root
[sudo] password for deven:
oot@ubuntu:/home/deven/lab1/Labsetup# ls -l catall
rwsr-xr-x 1 root deven 16184 Mar 28 02:33 catall
oot@ubuntu:/home/deven/lab1/Labsetup# sudo su sally
ally@ubuntu:/home/deven/lab1/Labsetup$ nano file2delete
ally@ubuntu:/home/deven/lab1/Labsetup$ sudo su root
[sudo] password for sally:
oot@ubuntu:/home/deven/lab1/Labsetup# nano file2delete.txt
```

If we run the program as a user without permission but still enter proper bash syntax it will execute

```
sally@ubuntu:/home/deven/lab1/Labsetup$ ls
a.out       catall.c         foo                 myenv    mylib.o
cap_leak.c  file             foo.c               myenv.c  myprintenv.c
catall      file2delete.txt  libmylib.so.1.0.1   mylib.c  myprog
sally@ubuntu:/home/deven/lab1/Labsetup$ ./catall
Please type a file name.
sally@ubuntu:/home/deven/lab1/Labsetup$ rm file2delete.txt
rm: remove write-protected regular file 'file2delete.txt'? yes
rm: cannot remove 'file2delete.txt': Permission denied
```

The permission was denied it maybe the case linux has a built in defense against this form of attack. I will repeat the same except system() will be repplaces with execve()

```
deven@ubuntu:~/lab1/Labsetup$ nano catall
deven@ubuntu:~/lab1/Labsetup$ nano catall.c
deven@ubuntu:~/lab1/Labsetup$ gcc catall.c -o catall
deven@ubuntu:~/lab1/Labsetup$ ./catall
Please type a file name.
deven@ubuntu:~/lab1/Labsetup$ rm file2delete.txt
rm: remove write-protected regular file 'file2delete.txt'? yes
deven@ubuntu:~/lab1/Labsetup$ ls
a.out       catall.c  foo.c              myenv.c  myprintenv.c  myprog.o    output1.txt
cap_leak.c  file      libmylib.so.1.0.1  mylib.c  myprog        mysystem    output2.txt
catall      foo       myenv              mylib.o  myprog.c      mysystem.c  output3.txt
deven@ubuntu:~/lab1/Labsetup$
```

This time the attack was successful. Although, I suspect it may be because I changed users. However, root is still the owner so I am unsure.

Task 9: Capability Leaking

The cap leak program has a vulnerability by allowing the user to access the shell through input. Beacause its in a program owned by root permissions are bypassed and anything a user enters can be used for a command. Here is an example of how I had it print “Hacked”

```
deven@ubuntu:~/lab1/Labsetup$ gcc cap_leak.c -o cap_leak
deven@ubuntu:~/lab1/Labsetup$ sudo su root
[sudo] password for deven:
root@ubuntu:/home/deven/lab1/Labsetup# sudo chown root cap_leak
root@ubuntu:/home/deven/lab1/Labsetup# sudo chmod 4755 cap_leak
root@ubuntu:/home/deven/lab1/Labsetup# sudo su deven
deven@ubuntu:~/lab1/Labsetup$ ./cap_leak
Cannot open /etc/zzz
deven@ubuntu:~/lab1/Labsetup$ sudo su root
[sudo] password for deven:
root@ubuntu:/home/deven/lab1/Labsetup# sudo chwon root cap_leak
sudo: chwon: command not found
root@ubuntu:/home/deven/lab1/Labsetup# sudo chown root cap_leak
root@ubuntu:/home/deven/lab1/Labsetup# chmod 4755 cap_leak
root@ubuntu:/home/deven/lab1/Labsetup# sudo su deven
deven@ubuntu:~/lab1/Labsetup$ ./cap_leak
Cannot open /etc/zzz
deven@ubuntu:~/lab1/Labsetup$ sudo su root
[sudo] password for deven:
root@ubuntu:/home/deven/lab1/Labsetup# sudo touch /etc/zzz
root@ubuntu:/home/deven/lab1/Labsetup# sudo chmod 644 /etc/zzz
root@ubuntu:/home/deven/lab1/Labsetup# sudo su deven
deven@ubuntu:~/lab1/Labsetup$ ./ cap_leak
bash: ./: Is a directory
deven@ubuntu:~/lab1/Labsetup$ ./cap_leak
fd is 3
$ echo "Hacked" >&3
$
$ cat /etc/zzz
Hacked
$
```